

The Intra-Plane Spectral Sweep Propagator

Directional-Residue Bloch Summation: Closed-Form Lateral Sweeps, Poisson Collapse, and the Kambe Equivalence

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Abstract

The Bloch-periodic Green’s tensor that couples a plane of identical scatterers at equal depth is conditionally convergent in the real-space lattice sum: the evanescent damping that makes the inter-plane propagator converge disappears when the source and receiver share a depth. The companion note on Ewald summation cures this by a Gaussian theta-split. Here we develop a different and complementary representation of the *same* object, the one that the thesis calls the spectral x/y -split and that the directional-sweep solver uses as its matrix–vector product. Each constituent plane wave is carried along whichever lateral axis has the larger receiver coordinate, and along that axis the Bloch lattice sum is an exact *geometric* series, resummed in closed form; the orthogonal lateral sum is collapsed by Poisson summation onto the reciprocal lattice; and the remaining depth integral is a one-dimensional quadrature. The net construction carries no spatial truncation in either lateral direction. We give the residue (angular-spectrum) representation of the scalar propagator and its general-depth extension, derive the geometric row identity that is the heart of the sweep, perform the Poisson collapse, and reduce the elastic tensor to the scalar lattice structure constants D_{qs} that the Kambe vector propagator is assembled from. A closing section records the damped-regime numerical validation: the directional-residue D_{qs} reproduces the Kambe addition-theorem sum, and the closed-form row reproduces the brute lattice sum to machine precision.

Companion series. *One of a companion series on elastic multiple scattering from cubic heterogeneities on a depth-stratified background, the T -matrix programme $T = T_0(I - G_0T_0)^{-1}$: (I) the stratified inter-site propagator, the same-depth convergence theory, and Ewald summation of the intra-plane coupling (`EwaldIntraPlanePropagator`); (II) the contrast-source volume integral equation and the CG-FFT grid T -matrix lineage (`ContrastSourceVIE`); (III) the directional partial-wave sweep matvec and the Krylov multiple-scattering solve (`DirectionalSweepSolver`). This note develops the directional-residue (spectral sweep) representation of the intra-plane propagator that underlies the matvec of (III) and is the spectral counterpart to the Ewald construction of (I).*

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1 The intra-plane object and why a directional representation

Fix a horizontal slowness through the Bloch wavevector $\mathbf{k}_\parallel$ and a planar Bravais lattice $\Lambda = \{\mathbf{R}_{ij} = (a_L i, a_L j, 0)\}$ of scatterer sites at a common depth. The quantity that closes the intra-plane multiple-scattering system is the Bloch-summed scalar Helmholtz propagator

$$G(\mathbf{r}; \kappa) = \sum_{\mathbf{R} \in \Lambda \setminus \{0\}} g(\mathbf{r} - \mathbf{R}; \kappa) e^{i\mathbf{k}_\parallel \cdot \mathbf{R}}, \quad g(\mathbf{d}; \kappa) = \frac{e^{i\kappa|\mathbf{d}|}}{4\pi|\mathbf{d}|}, \quad (1)$$

evaluated for $\mathbf{r}$ in the source plane (the elastic vector tensor reduces to two such scalar sums, at $\kappa = \kappa_P$ and $\kappa = \kappa_S$; see §5). The self term $\mathbf{R} = 0$ is excluded; it is restored separately as the single-site interaction.

For sites at *different* depths the plane-wave (partial-wave) form of g carries an evanescent factor $e^{-\text{Im}(k_z)|\Delta z|}$ whose rate grows with the depth separation, and the lattice sum converges geometrically. At equal depth, $\Delta z = 0$, that factor is unity and (1) is only conditionally convergent: $g \sim 1/|\mathbf{R}|$ while the lattice has $\mathcal{O}(R)$ sites per shell. Companion note (I) records this breakdown and resolves it by Ewald summation. The present note resolves the *same* sum by a directional route that keeps the physics of a propagating partial wave manifest.

The idea is that each constituent plane wave should be carried along the axis in which it is naturally damped, and that along that axis the periodicity of Λ turns the lattice sum into a geometric series. We therefore (i) write g as an angular spectrum with one lateral axis singled out as the *swept* axis (§2); (ii) resum the swept-axis lattice sum in closed form (§3); (iii) collapse the orthogonal lateral sum by Poisson summation, leaving a one-dimensional depth quadrature (§4); and (iv) project onto multipoles to recover the Kambe structure constants (§5).

2 The directional-residue representation of the propagator

The starting point is the Weyl angular-spectrum identity for the free-space Helmholtz Green's function. Singling out a lateral “swept” axis s (one of x or y) and writing the two transverse wavenumbers as (k_a, k_b) ,

$$g(\mathbf{d}; \kappa) = \frac{i}{8\pi^2} \iint_{\mathbb{R}^2} \frac{e^{ik_\perp|d_s|}}{k_\perp} e^{i(k_a d_a + k_b d_b)} dk_a dk_b, \quad k_\perp = \sqrt{\kappa^2 - k_a^2 - k_b^2}, \quad (2)$$

with the branch $\text{Im } k_\perp \geq 0$ fixed so that $e^{ik_\perp|d_s|}$ decays. The role of the depth coordinate in the usual Weyl integral is here played by the swept lateral coordinate d_s : the integrand decays like $e^{-\text{Im}(k_\perp)|d_s|}$, so the representation converges fastest when $|d_s|$ is large. We therefore choose the swept axis per displacement as the one with the larger lateral separation,

$$s = \arg \max(|d_x|, |d_y|), \quad (3)$$

the *dual sweep*: it guarantees a non-vanishing evanescent rate for every off-axis site and is what fixes the near-axis convergence of the reciprocal sum in §4.

General depth. The intra-plane projection in §5 needs the field on a small sphere about a site, hence at $d_z \neq 0$. The representation (2) extends to general depth simply by carrying the depth phase in the remaining transverse integral: with $b \equiv z$,

$$g(\mathbf{d}; \kappa) = \frac{i}{8\pi^2} \int dk_z e^{ik_z d_z} \int dk_a \frac{e^{ik_\perp |d_s|}}{k_\perp} e^{ik_a d_a}, \quad k_\perp = \sqrt{\kappa^2 - k_a^2 - k_z^2}, \quad (4)$$

and for $|d_s| > 0$ the evanescent factor enforces absolute convergence of the (k_a, k_z) integral regardless of d_z .

Quadrature of the propagating disk. The integrand of (2) is near-singular on the circle $k_a^2 + k_b^2 = \text{Re}(\kappa^2)$, where $k_\perp \rightarrow 0$ and the kernel $1/k_\perp$ diverges (the damping $\text{Im} \kappa > 0$ keeps it finite but sharply peaked). This propagating disk must be *resolved*, not merely integrated to high polynomial order. A dense *uniform* grid spreads its nodes evenly across the disk and resolves the ring; a Gauss–Legendre rule of the same node count clusters its nodes near the interval ends and systematically under-resolves the ring, biasing the result. This is the single most important practical point in the construction and is illustrated in Figure 1.

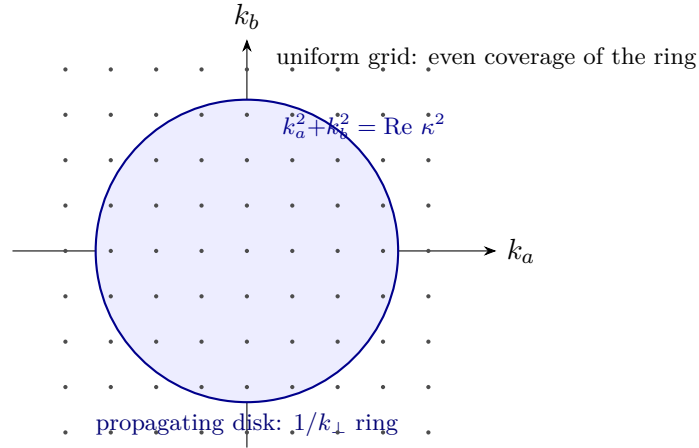


Figure 1: Quadrature of the angular-spectrum integral (2). The kernel $1/k_\perp$ is near-singular on the propagating circle $k_a^2 + k_b^2 = \text{Re} \kappa^2$ (the damping $\text{Im} \kappa > 0$ keeps it finite). A dense *uniform* grid (dots) covers the ring evenly and resolves it; a Gauss–Legendre rule of equal node count clusters toward the interval ends and under-resolves the ring, biasing the lattice sum.

3 The lateral sweep: geometric resummation

Insert the swept-axis representation into the Bloch sum and isolate the one-dimensional sublattice along the swept axis. With site index $i \in \mathbb{Z}$ at swept-axis position $a_L i$, swept-axis receiver coordinate r_s , and Bloch phase $k_{\parallel,s}$, the swept-axis sum is

$$S(k_\perp, r_s) = \sum_{i \in \mathbb{Z}} e^{ik_\perp |r_s - a_L i|} e^{ik_{\parallel,s} a_L i}. \quad (5)$$

The absolute value in the exponent is the only obstruction to this being a plain geometric series, and it changes sign exactly once, at $i_0 = \lfloor r_s / a_L \rfloor$. Splitting the sum there removes the modulus:

$$\begin{aligned} i \leq i_0 : \quad r_s - a_L i \geq 0 &\Rightarrow e^{ik_\perp r_s} p^i, \quad p = e^{i(k_{\parallel,s} - k_\perp) a_L}, \\ i > i_0 : \quad r_s - a_L i < 0 &\Rightarrow e^{-ik_\perp r_s} q^i, \quad q = e^{i(k_{\parallel,s} + k_\perp) a_L}. \end{aligned} \quad (6)$$

With the branch $\text{Im } k_\perp > 0$ of §2 one has $|p| = e^{\text{Im}(k_\perp)a_L} > 1$, so the left tail $i \rightarrow -\infty$ converges, and $|q| = e^{-\text{Im}(k_\perp)a_L} < 1$, so the right tail $i \rightarrow +\infty$ converges. Each side is a geometric series, and (5) sums in closed form:

$$S(k_\perp, r_s) = e^{ik_\perp r_s} \frac{p^{i_0+1}}{p-1} + e^{-ik_\perp r_s} \frac{q^{i_0+1}}{1-q} \quad (7)$$

This is the thesis cumulative-phase sweep (and the FFTProp row identity): the phase of a propagating partial wave accumulates *linearly* from site to site, so the contribution of an entire row of the lattice is a single closed-form expression rather than a truncated sum. The geometry of the split is shown in Figure 2.

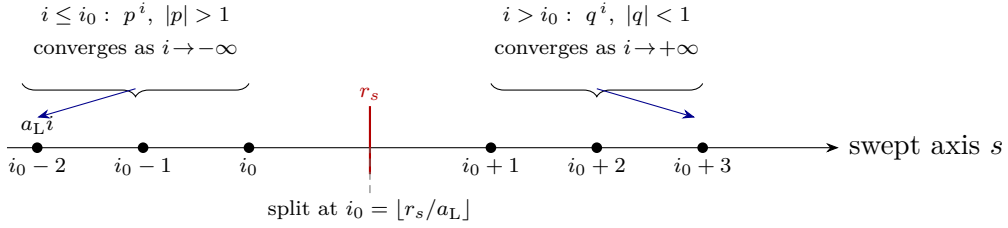


Figure 2: The geometric row identity (7). The modulus $|r_s - a_L i|$ in (5) changes sign once, at $i_0 = \lfloor r_s/a_L \rfloor$. To the left the ratio p has $|p| > 1$ and the sum to $-\infty$ converges; to the right $|q| < 1$ and the sum to $+\infty$ converges. Each side is summed in closed form, so an entire lattice row is evaluated without truncation.

4 Poisson collapse and the depth quadrature

The orthogonal lateral direction (the “reciprocal” axis, with receiver coordinate r_r and Bloch phase $k_{\parallel,r}$) is still a sum over lattice index j . Rather than sum it in real space, we apply Poisson summation to the j -sum, which collapses the corresponding transverse integral in (4) onto the reciprocal lattice:

$$\sum_{j \in \mathbb{Z}} e^{ik_a(r_r - a_L j)} e^{ik_{\parallel,r} a_L j} = \frac{2\pi}{a_L} \sum_{n \in \mathbb{Z}} \delta(k_a - k_{r,n}) e^{ik_{r,n} r_r}, \quad k_{r,n} = k_{\parallel,r} + \frac{2\pi n}{a_L}. \quad (8)$$

The transverse k_a -integral is then done by the delta functions, fixing it to the reciprocal-lattice values $k_{r,n}$. Combining the prefactor $i/8\pi^2$ of (4) with the factor $2\pi/a_L$ leaves $i/(4\pi a_L)$, and the resummed intra-plane propagator is

$$G(\mathbf{r}; \kappa) = \frac{i}{4\pi a_L} \sum_{n \in \mathbb{Z}} e^{ik_{r,n} r_r} \int dk_z e^{ik_z r_z} \frac{S(k_\perp, r_s)}{k_\perp} - g(\mathbf{r}; \kappa), \quad k_\perp = \sqrt{\kappa^2 - k_{r,n}^2 - k_z^2}, \quad (9)$$

where S is the closed-form row sum (7), the final $g(\mathbf{r})$ subtracts the self ($i = j = 0$) site that (1) excludes, and the swept/reciprocal assignment (s, r) follows the dual-sweep rule (3).

The three lattice directions are thus handled by three different and each exact mechanisms: the swept lateral axis in closed form, the orthogonal lateral axis by a fast reciprocal sum, and the depth by a one-dimensional quadrature. The reciprocal sum over n converges geometrically because for $|k_{r,n}| > |\kappa|$ the wavenumber $k_\perp$ is evanescent and $S/k_\perp$ decays like $e^{-|k_{r,n}| |r_s|}$; the dual-sweep choice $|r_s| \geq |r_r|$ maximises this decay rate. No spatial truncation is taken in either lateral direction. The structure is summarised in Figure 3.

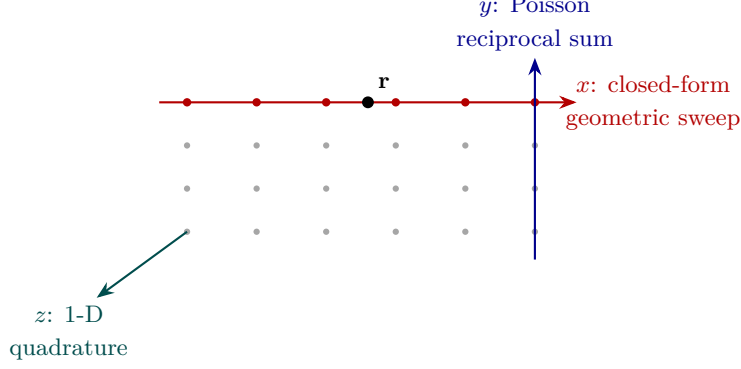


Figure 3: The spectral sweep over the planar lattice (9). The swept lateral axis (here x , red) is resummed in closed form (7); the orthogonal lateral axis (y , blue) is a fast reciprocal sum after Poisson collapse (8); the depth (z , out of plane) is a one-dimensional quadrature. The swept axis is chosen per receiver as $\arg \max(|r_x|, |r_y|)$.

5 From scalar to vector: the Kambe structure constants

The elastic Bloch propagator $\mathbf{G}_0^{\text{vec}}(\mathbf{k}_{\parallel})$ is not an independent object: an exact operator identity reduces it to the two scalar Helmholtz lattice sums (1) at the compressional and shear wavenumbers, κ_P and κ_S , dressed with standard angular (vector spherical harmonic) coefficients. The irreducible content is therefore the set of scalar *lattice structure constants* D_{qs} , the multipole coefficients of the periodic field about a site. Establishing the spectral sweep at the scalar level is hence sufficient: the vector propagator follows by contraction.

We extract D_{qs} by projecting the resummed field (9) onto spherical harmonics over a small sphere $|\mathbf{r}| = \rho_0$ about a site,

$$D_{qs}^{\text{sweep}} = \frac{1}{i\kappa j_q(\kappa\rho_0)} \oint G(\rho_0\hat{\mathbf{n}}; \kappa) \overline{Y_q^s(\hat{\mathbf{n}})} d\Omega, \quad (10)$$

which is to be compared with Kambe's addition-theorem evaluation of the same constants as a direct real-space sum of outgoing spherical waves,

$$D_{qs}^{\text{Kambe}} = \sum_{\mathbf{R} \in \Lambda \setminus \{0\}} h_q^{(1)}(\kappa|\mathbf{R}|) \overline{Y_q^s(\frac{\pi}{2}, \phi_{\mathbf{R}})} e^{i\mathbf{k}_{\parallel} \cdot \mathbf{R}}. \quad (11)$$

Both sums converge absolutely in the damped regime $\text{Im } \kappa > 0$. The vector projectors (regular/outgoing multipoles $\mathbf{L}, \mathbf{M}, \mathbf{N}$ in the Cruzan/Stein normalisation, built from the radial and tangential vector spherical harmonics) are ported and self-tested separately; once D_{qs} agrees, the vector match is redundant. Figure 4 places the spectral sweep among the three representations of the same intra-plane propagator and the Ewald route of companion (I).

6 Numerical validation (damped regime)

The construction was validated as a chain of tracer-bullet checks in the damped regime $\text{Im } \omega > 0$, where both the real-space lattice sum and its spectral resummation converge absolutely; the damping η (with $\omega = 2\pi(1 + i\eta)$) is a genuine physical attenuation, chosen per check for fast convergence. Each check isolates one link in the chain of Figure 4. Background $\alpha = 5$, $\beta = 3$; lattice pitch $a_L = 2$; Bloch vector $\mathbf{k}_{\parallel} = (0.2, 0.1)$.

The two extremes of Table 1 carry the most information. The row identity TB3a is exact to machine precision (2×10^{-15}): the closed form (7) is the brute lattice sum, with no approximation beyond floating point. At the other end, TB2b confirms the substantive cross-code statement,

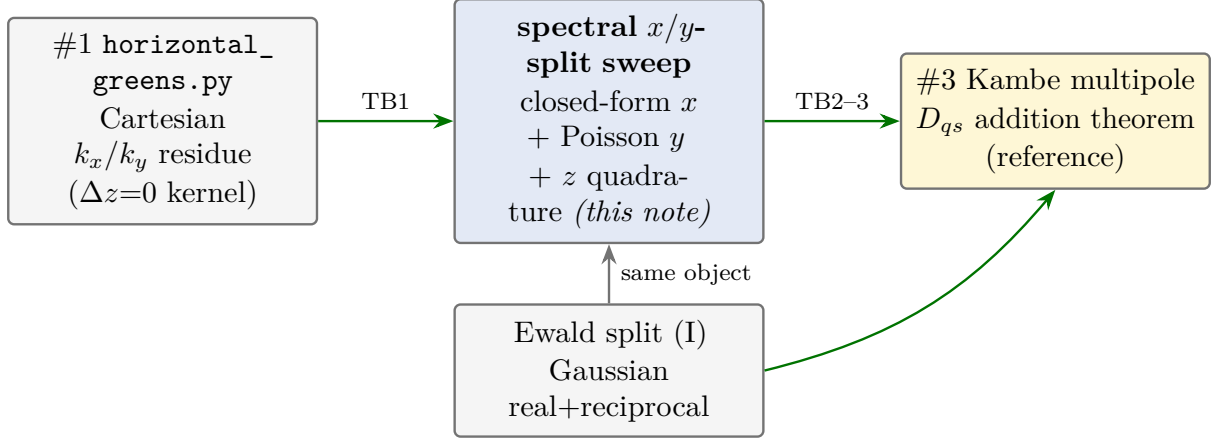


Figure 4: The intra-plane propagator in three representations and two summation routes. The directional x/y -split sweep (centre, this note) is the spectral bridge: it extends the equal-depth Cartesian residue kernel of `horizontal_greens.py` (#1) to general depth and to the infinite lattice, and reproduces the Kambe multipole structure constants D_{qs} (#3, the reference). The Ewald split of companion (I) is an alternative summation of the same object. Green arrows are verified numerical equalities (§6).

Table 1: Damped-regime validation of the intra-plane spectral sweep. Each row reports the relative agreement of one link in the chain `horizontal_greens` $\rightarrow$ residue $\rightarrow$ sweep $\rightarrow$ Kambe.

Check	What is compared	Statement	Residual
TB1	residue kernel vs. exact, in-plane	$G_{ik}^{\text{res}}(\mathbf{R}) = G_{ik}^{\text{exact}}(\mathbf{R})$ at lattice vectors	2.9×10^{-5}
TB2a	general- z residue vs. exact	(4) off-plane	1.3×10^{-4}
TB2b	structure constants, residue vs. Kambe	$D_{qs}^{\text{sweep}} = D_{qs}^{\text{Kambe}}, \kappa_P/\kappa_S$	$2.3 \times 10^{-4} / 2.5 \times 10^{-7}$
TB3a	row resummation vs. brute	(7) = truncated (5), $M=4000$	2×10^{-15}
TB2c	vector $\mathbf{L}/\mathbf{M}/\mathbf{N}$ projector	self-consistency (reconstruct own coefficients)	6×10^{-16}

that the spectral-sweep structure constants equal the Kambe addition-theorem constants; the residual there is set by the angular quadrature on the projection sphere and by the residue quadrature of the propagating disk (Figure 1), and tightens with node count. An intermediate gate (not shown) confirms that projecting the *exact* lattice field with the same projector returns the Kambe constants to 10^{-13} , so the small TB2b residual is quadrature, not a representation error. The κ_S residual is far smaller than κ_P because the larger shear wavenumber places the propagating disk where the fixed uniform grid resolves it more finely.

The full sweep-resummed field of (9) (TB3b) reproduces both the brute lattice sum and the Kambe constants within tolerance, confirming that the closed-form row, the Poisson reciprocal sum, and the depth quadrature compose into the infinite-lattice limit that Kambe converges to.

7 Relationship to the companion constructions

The spectral sweep and the Ewald split of companion (I) compute the same conditionally convergent object (1) by different analytic continuations. Ewald splits each term into a Gaussian-damped real-space part and a Gaussian-damped reciprocal-space part, both absolutely convergent for any geometry; it is the method of choice for a fully irregular evaluation and is agnostic to direction. The spectral sweep instead exploits the one-dimensional periodicity of a lattice *row*:

along the swept axis the sum is not merely accelerated but eliminated, replaced by the closed form (7), and only the orthogonal lateral direction carries a (reciprocal) sum. This is exactly the structure that makes it a matrix–vector product in companion (III): the directional partial-wave sweep there is the operator version of (9), applied across the voxel grid, with the depth handled by the Kennett recursion between planes rather than by quadrature.

In the language of the three representations (Figure 4), the sweep is the bridge that promotes the equal-depth Cartesian residue kernel of `horizontal_greens.py` to the general-depth infinite-lattice propagator, and validates it against the Kambe multipole reference without ever forming a truncated real-space lattice sum.